

DS Unit 5 Question Bank — Sorting, Searching & Hashing

Data Structures (DS) · Unit 5 · Sorting, Searching & Hashing · 2025

Important and frequently asked questions, grouped by marks.

Part A — Short answer (2 marks)

1. What is meant by Binary Search? Discuss its advantages and limitations.
2. Explain Quick Sort with a neat diagram and a suitable example.
3. Write short notes on Merge Sort.
4. Write short notes on Heap Sort.
5. What is meant by Hash Functions? Discuss its advantages and limitations.
6. Define Collision Resolution. Derive or justify the key properties involved.

Part B — Long answer (10 marks)

1. Compare Binary Search with Merge Sort. When would you prefer one over the other?
2. Compare Quick Sort with Binary Search. When would you prefer one over the other?
3. Write short notes on Merge Sort.
4. Write short notes on Heap Sort.
5. Illustrate Hash Functions with a worked example.
6. Discuss the role of Collision Resolution in Sorting, Searching & Hashing.
7. Compare Binary Search with Collision Resolution. When would you prefer one over the other?
8. Illustrate Quick Sort with a worked example.
9. Define Merge Sort. Derive or justify the key properties involved.
10. Illustrate Heap Sort with a worked example.
11. What is meant by Hash Functions? Discuss its advantages and limitations.
12. Explain Collision Resolution with a neat diagram and a suitable example.